WORKED SOLUTION

STEP II 1987, Question 4

Pure • Complex Numbers and Geometry

STEP II 1987, Question 4

Read the question before you go on. It is in the STEP Support Programme's question database:

step.maths.org/questions/87-s2-q4

The key idea

Multiplying by $e^{i\theta}$ is a rotation, and an equilateral triangle is exactly what you get when one side is rotated through 60° about a vertex. That turns the geometry into a linear equation in α, β, γ. Allowing either orientation means *one of two* linear expressions vanishes, and their product is the symmetric quadratic in the question. Since it is symmetric, it can be written in terms of the coefficients of the cubic.

Solution

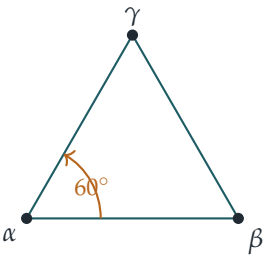
Throughout, $\omega = e^{2\pi i/3}$, so $\omega^3 = 1$, $\bar{\omega} = \omega^2$ and $1 + \omega + \omega^2 = 0$.

Multiplication by $e^{i\theta}$

If $z = re^{i\phi}$ then $ze^{i\theta} = re^{i(\phi+\theta)}$: the modulus is unchanged and the argument goes up by θ. So the point $ze^{i\theta}$ is the image of the point z under an anticlockwise rotation through angle θ about the origin.

More generally, $\gamma - \alpha = e^{i\theta}(\beta - \alpha)$ says that the vector from α to γ is the vector from α to β rotated anticlockwise through θ; that is, γ is the image of β under rotation through θ about α.

Anticlockwise equilateral triangles



Distinct points α, β, γ form an equilateral triangle in anticlockwise order exactly when γ is obtained from β by rotating through $\frac{\pi}{3}$ anticlockwise about α, i.e. when

$$\gamma - \alpha = e^{i\pi/3}(\beta - \alpha).$$

(Necessity: in such a triangle $|\alpha\gamma| = |\alpha\beta|$ and the angle at α is $\frac{\pi}{3}$, measured anticlockwise from $\alpha\beta$ to $\alpha\gamma$. Sufficiency: the equation gives $|\alpha\gamma| = |\alpha\beta|$ with angle $\frac{\pi}{3}$ between them, so the triangle is isosceles with apex angle $60°$, hence equilateral, and the orientation is anticlockwise.)

It is neater to write this symmetrically. Since $e^{i\pi/3} = -\omega^2$ (because $-\omega^2 = e^{i\pi}e^{4\pi i/3} = e^{i\pi/3}$, using $e^{7\pi i/3} = e^{i\pi/3}$), the condition becomes

$$\begin{aligned} & & \gamma - \alpha &= -\omega^2\beta + \omega^2\alpha \\ \iff & & \gamma + \omega^2\beta - (1+\omega^2)\alpha &= 0 \\ & \iff & \gamma + \omega^2\beta + \omega\alpha &= 0, \end{aligned}$$

using $1 + \omega^2 = -\omega$. Multiplying by ω^2 (which is non-zero, so this is reversible) gives the equivalent condition

$$\alpha + \omega\beta + \omega^2\gamma = 0.$$

A quick sanity check: $1, \omega, \omega^2$ are anticlockwise vertices of an equilateral triangle, and $1 + \omega \cdot \omega + \omega^2 \cdot \omega^2 = 1 + \omega^2 + \omega = 0$.

Either orientation

Any three distinct points forming an equilateral triangle are anticlockwise either in the order α, β, γ or in the order α, γ, β (cyclic re-orderings do not change the orientation). By the previous part, for distinct α, β, γ,

$$\begin{aligned} \text{equilateral} &\iff \alpha + \omega\beta + \omega^2\gamma = 0 \ \text{ or } \ \alpha + \omega\gamma + \omega^2\beta = 0 \\ &\iff (\alpha + \omega\beta + \omega^2\gamma)(\alpha + \omega^2\beta + \omega\gamma) = 0. \end{aligned}$$

Ask yourself How do we see that this product is the expression we want?

Just expand it. The squared terms give $\alpha^2+\omega^3\beta^2+\omega^3\gamma^2=\alpha^2+\beta^2+\gamma^2$. The cross terms are

$$\begin{aligned}\alpha\beta(\omega^2+\omega)+\alpha\gamma(\omega+\omega^2)+\beta\gamma(\omega^2+\omega^4) &= \alpha\beta(\omega+\omega^2)+\gamma\alpha(\omega+\omega^2)+\beta\gamma(\omega^2+\omega)\\ &= -\alpha\beta-\gamma\alpha-\beta\gamma,\end{aligned}$$

using $\omega^4=\omega$ and $\omega+\omega^2=-1$. Hence

$$(\alpha+\omega\beta+\omega^2\gamma)(\alpha+\omega^2\beta+\omega\gamma)=\alpha^2+\beta^2+\gamma^2-\beta\gamma-\gamma\alpha-\alpha\beta,$$

and so the distinct points α,β,γ form an equilateral triangle (in some order) if and only if

$$\alpha^2+\beta^2+\gamma^2-\beta\gamma-\gamma\alpha-\alpha\beta=0,$$

as required.

It is worth noting what happens without distinctness: if, say, $\alpha=\beta$, the expression is $(\alpha-\gamma)^2$, which vanishes only when $\gamma=\alpha$ too. So the expression is zero precisely when the points form an equilateral triangle *or all three coincide*. We need this below.

The cubic

Let the roots of $z^3+az^2+bz+c=0$ be α,β,γ. Then

$$\alpha+\beta+\gamma=-a,\qquad \beta\gamma+\gamma\alpha+\alpha\beta=b,\qquad \alpha\beta\gamma=-c.$$

The expression is symmetric, so we write it in terms of these:

$$\begin{aligned}\alpha^2+\beta^2+\gamma^2-(\beta\gamma+\gamma\alpha+\alpha\beta) &= (\alpha+\beta+\gamma)^2-3(\beta\gamma+\gamma\alpha+\alpha\beta)\\ &= a^2-3b.\end{aligned}$$

By the remark above, $a^2=3b$ exactly when the roots form an equilateral triangle or are all equal. All three roots equal t means $z^3+az^2+bz+c=(z-t)^3$, so $a=-3t$, $b=3t^2$, $c=-t^3$; that is, $b=\frac{a^2}{3}$ and $c=\frac{a^3}{27}$. So the roots lie at the vertices of a (genuine) equilateral triangle if and only if

$$\boxed{a^2=3b \quad\text{and}\quad 27c\neq a^3}$$

If one is willing to count three coincident points as a degenerate equilateral triangle, the condition is simply $a^2=3b$.

There is a nice way to see this directly. Substituting $z=w-\frac{a}{3}$ removes the z^2 term; when $a^2=3b$ the linear term vanishes too and the equation becomes $w^3=\frac{a^3}{27}-c$. Its roots are the three cube roots of a non-zero number, which form an equilateral triangle centred at $w=0$, i.e. at $z=-\frac{a}{3}$, the centroid of the roots.

Checks

- The roots of $z^3-1=0$ ($a=b=0$, $c=-1$) are $1,\omega,\omega^2$: equilateral, and indeed $a^2=0=3b$ and $27c=-27\neq 0$.

- The roots of $z^3 - z = 0$ are $0, \pm 1$, collinear; here $a^2 - 3b = 3 \neq 0$.
- Numerically, for 1000 random triangles α, β, $\gamma = \alpha + e^{i\pi/3}(\beta - \alpha)$, we checked that the triangle is anticlockwise and equilateral, that $\alpha + \omega\beta + \omega^2\gamma$ and the quadratic expression vanish (in every ordering), and that the product identity and $a^2 - 3b$ formula hold for random complex numbers to within 10^{-8}.

Common slips

- Getting the sign or orientation wrong: $e^{i\pi/3}$ is $-\omega^2$, not ω; rotating by ω (that is, $120°$) about a vertex does not give an equilateral triangle.
- Forgetting the triple-root case: $a^2 = 3b$ alone also allows $z^3 + az^2 + bz + c = (z + \frac{a}{3})^3$, where the three "vertices" coincide.

Beyond STEP

The condition $\alpha + \omega\beta + \omega^2\gamma = 0$ is one component of a discrete Fourier transform of the triple (α, β, γ), and the trick of removing the z^2 term is the first step of solving the cubic by radicals, which leads into Galois theory. Geometrically, the result says the roots are equilateral exactly when the derivative $3z^2 + 2az + b$ has a double root, at the centroid; this is a special case of Marden's theorem about the roots of the derivative. Needham's *Visual Complex Analysis* is a beautiful place to explore the geometry of complex multiplication further.

Beyond STEP worked solutions: https://beyondstep.keithmcnulty.org/. Solution text CC0 1.0; the question itself is © Cambridge University Press & Assessment and is not reproduced here.